

Multiple Choice (10 marks)

1. Which of the following most likely explains why Venus does not have a strong magnetic field?
 - (a) Its rotation is too slow.
 - (b) It has too thick an atmosphere.
 - (c) It is too large.
 - (d) It does not have a metallic core.
2. Life on Earth originated from
 - (a) We don't know for sure.
 - (b) aliens (panspermia).
 - (c) comets.
 - (d) the Earth's oceans.
3. How do astronomers think Jupiter generates its internal heat?
 - (a) nuclear fusion in the core
 - (b) by contracting changing gravitational potential energy into thermal energy
 - (c) internal friction due to its high rotation rate
 - (d) chemical processes
4. What type of radiation causes a black hole to evaporate over time?
 - (a) Schwarzschild radiation
 - (b) Planck radiation
 - (c) Kolmogorov radiation
 - (d) Hawking radiation
5. Which planet(s) in the solar system has/have active plate tectonics?
 - (a) Mars
 - (b) Venus
 - (c) Earth
 - (d) Mars and Earth

True / False (10 marks)

Answer True or False

6. True or False: The two moons of Mars are named Phobos and Deimos.
7. True or False: Mars lacks an ozone layer, which is essential for a strong greenhouse effect to occur on a planet.
8. True or False: All forms of life require the ability to make energy from sunlight or consume organisms that do so for survival.
9. True or False: The Andromeda Galaxy is located about 1.7 million light years from Earth, making it one of the closest galaxies to us.
10. True or False: Saturn and Jupiter do not both exhibit equatorial wind speeds exceeding 900 miles per hour in their atmospheres.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the current challenges faced by the Mars Exploration Rovers, Opportunity and Spirit, and analyze how these challenges impact their scientific missions.
12. Discuss the significance of the Arecibo Telescope in the field of radio astronomy, focusing on its design, capabilities, and contributions to our understanding of the universe.

End of Paper